

Polynomial WS 1

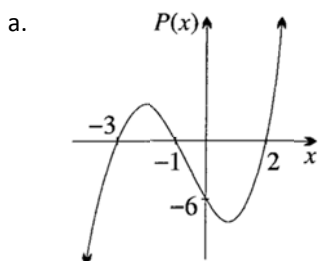
1. a. Find the quotient and remainder when $x^4 - 2x^3 + x^2 - 5x + 7$ is divided by $x^2 + x - 1$.
b. Find a and b if $x^4 - 2x^3 + x^2 + ax + b$ is exactly divisible by $x^2 + x - 1$.
c. Hence factor $x^4 - 2x^3 + x^2 + 8x - 5$.
2. a. Use long division to divide the polynomial $f(x) = x^4 - x^3 + x^2 - x + 1$ by the polynomial $d(x) = x^2 + 4$. Express your answer in the form $f(x) = d(x)q(x) + r(x)$.
b. Hence find the values of a and b such that $x^4 - x^3 + x^2 + ax + b$ is divisible by $x^2 + 4$.
c. Hence factor $x^4 - x^3 + x^2 - 4x - 12$.
3. If $x^4 - 2x^3 - 20x^2 + ax + b$ is exactly divisible by $x^2 - 5x + 2$, find a and b .
4. a. Find k , if $x - 1$ is a factor of $P(x) = x^3 - 3x^2 + kx - 2$.
b. Find m , if -2 is a zero of the function $F(x) = x^3 + mx^2 - 3x + 4$.
c. When the polynomial $P(x) = 2x^3 - x^2 + px - 1$ is divided by $x - 3$, the remainder is 2. Find p .
d. For what value of a is $3x^4 + ax^2 - 2$ divisible by $x + 1$?
5. a. Show that $P(x) = x^3 - 8x^2 + 9x + 18$ is divisible by $x - 3$ and $x + 1$.
b. By considering the leading term and constant term, express $P(x)$ in terms of three linear factors and hence solve $P(x) \geq 0$.
6. a. Show that $P(x) = x^3 - 8x^2 + 9x + 18$ is divisible by $x - 3$ and $x + 1$.
b. By considering the leading term and constant term, express $P(x)$ in terms of three linear factors and hence solve $P(x) \leq 0$.
7. Factor each of the following polynomials and sketch a graph, indicating all intercepts with the axes. You do not need to find any other turning points.
 - a. $P(x) = x^3 + 2x^2 - 5x - 6$
 - b. $P(x) = x^3 + 3x^2 - 25x + 21$
 - c. $P(x) = -x^3 + x^2 + 5x + 3$
 - d. $P(x) = x^4 - x^3 - 19x^2 - 11x + 30$
 - e. $P(x) = 2x^3 + 11x^2 + 10x - 8$
 - f. $P(x) = 3x^4 + 4x^3 - 35x^2 - 12x$
8. a. If $P(x) = 2x^3 + x^2 - 13x + 6$, evaluate $P\left(\frac{1}{2}\right)$. Use long division to express $P(x)$ in factored form.
b. If $P(x) = 6x^3 + x^2 - 5x - 2$, evaluate $P\left(-\frac{2}{3}\right)$. Express $P(x)$ in factored form.
9. a. Factor each of the polynomials $P(x) = x^3 - 3x^2 + 4$, $Q(x) = x^3 + 2x^2 - 5x - 6$ and $R(x) = x^3 - 3x - 2$.
b. Hence determine the highest monic common factor of $P(x)$, $Q(x)$ and $R(x)$.
c. What is the monic polynomial of least degree that is exactly divisible by $P(x)$, $Q(x)$ and $R(x)$? Write the answer in factored form.
10. At time t , the position of a particle moving along the x -axis is given by the equation $x = t^4 - \frac{17}{3}t^3 + 8t^2 - 3t + 5$. Find the times at which the particle is stationary.
11. a. The polynomial $2x^3 - x^2 + ax + b$ has a remainder of 16 on division by $x - 1$ and a remainder of -17 on division by $x + 2$. Find a and b .
b. A polynomial is given by $P(x) = x^3 + ax^2 + bx - 18$. Find a and b , if $x + 2$ is a factor and -24 is the remainder when $P(x)$ is divided by $x - 1$.
c. $P(x)$ is an odd polynomial of degree 3. It has $x + 4$ as a factor, and when it is divided by $x - 3$ the remainder is 21. Find $P(x)$.
d. Find p such that $x - p$ is a factor of $4x^3 - (10p - 1)x^2 + (6p^2 - 5)x + 6$.

12. a. The polynomial $P(x)$ is divided by $(x - 1)(x + 2)$. Find the remainder, given that $P(1) = 2$ and $P(-2) = 5$. [HINT: The remainder may have degree 1.]
 b. The polynomial $U(x)$ is divided by $(x + 4)(x - 3)$. Find the remainder, given that $U(-4) = 11$ and $U(3) = -3$.
13. a. The polynomial $P(x) = x^3 + bx^2 + cx + d$ has zeroes at 0, 3 and -3 . Find b, c and d .
 b. Sketch a graph of $y = P(x)$.
 c. Hence solve $\frac{x^2 - 9}{x} > 0$.
14. a. Show that the equation of the normal to the curve $x^2 = 4y$ at the point $(2t, t^2)$ is $x + ty - 2t - t^3 = 0$.
 b. If the normal passes through the point $(-2, 5)$, find the value of t .
15. a. Is either $x + 1$ or $x - 1$ a factor of $x^n + 1$, where n is a positive integer?
 b. Is either $x + a$ or $x - a$ a factor of $x^n + a^n$, where n is a positive integer?
16. When a polynomial is divided by $(x - 1)(x + 3)$, the remainder is $2x - 1$.
 a. Express this in terms of a division identity statement.
 b. Hence, by evaluating $P(1)$, find the remainder when the polynomial is divided by $x - 1$.
17. a. When a polynomial is divided by $(2x + 1)(x - 3)$, the remainder is $3x - 1$. What is the remainder when the polynomial is divided by $2x + 1$?
 b. When $x^5 + 3x^3 + ax + b$ is divided by $x^2 - 1$, the remainder is $2x - 7$. Find a and b .
 c. When a polynomial $P(x)$ is divided by $x^2 - 5$, the remainder is $x + 4$. Find the remainder when $P(x) + P(-x)$ is divided by $x^2 - 5$. [HINT: Write down the division identity statement.]
18. Each term of an arithmetic sequence $a, a + d, a + 2d, \dots$ is added to the corresponding term of the geometric sequence $b, ba, ba^2, \dots$ to form a third sequence S , whose first three terms are $-1, -2$ and 6 . (Note that the common ratio of the geometric sequence is equal to the first term of the arithmetic sequence.)
 a. Show that $a^3 - a^2 - a + 10 = 0$
 b. Find a , given that a is real.
 c. Hence show that the n th term of S is given by $T_n = 2x - 4 + (-2)^{n-1}$.
19. a. The polynomial $(a - 2)x^2 + (1 - 3b)x + (5 - 2c)$ has three zeroes. What are the values of a, b and c ?
 b. The polynomial $(a + 1)x^3 + (b - 3)x^2 + (2x - 1)x + (5 - 4d)$ has four zeroes. What are the values of a, b, c and d ?
20. a. If $3x^2 - 4x + 7 \equiv a(x + 2)^2 + b(x + 2) + c$, find a, b and c .
 b. If $2x^3 - 8x^2 + 3x - 4 \equiv a(x - 1)^3 + b(x - 1)^2 + c(x - 1) + d$, find a, b, c and d .
 c. Use similar methods to express $x^3 + 2x^2 - 3x + 1$ as a polynomial in $(x + 1)$.
 d. If the polynomials $2x^2 + 4x + 4$ and $a(x + 1)^2 + b(x + 2)^2 + c(x + 3)^2$ are equal for three values of x , find a, b and c .
21. a. Express $2x^2 + 3x - 6$ in the form $a(x + 1)^2 + b(x + 1) + c$.
 b. Find a, b and c if $2x^2 + 4x + 5 \equiv a(x - 3)^2 + b(x - 3) + c$.
 c. Express $2x^2 - 5x + 3$ in the form $a(x - 2)^2 + b(x - 2) + c$.
22. a. Express x^3 in the form $a(x + 1)^2 + b(x + 1) + c$
 b. Express n^2 as a quadratic in $(n - 4)$.
 c. Express x^2 as a quadratic in $(x + 2)$.
23. a. If $2x^2 - x - 1 = a(x - b)(x - c)$ for all real values of x , find a, b and c .
 b. Express m^2 in the form $a(m - 1)^2 + b(m - 2)^2 + c(m - 3)^2$.

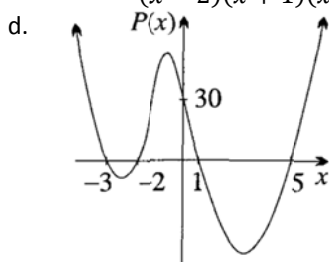
24. a. A polynomial of degree 3 has a double zero at 2. When $x = 1$ it takes the value 6 and when $x = 3$ it takes the value 8. Find the polynomial.
 b. Two zeroes of a polynomial of degree 3 are 1 and -3 . When $x = 2$ it takes the value -15 and when $x = -1$ it takes the value 36. Find the polynomial.
22. Show that $x^2 - 3x + 2$ is a factor of $P(x) = x^n(2^m - 1) + x^m(1 - 2^n) + (2^n - 2^m)$, where m and n are positive integers.

Answer:

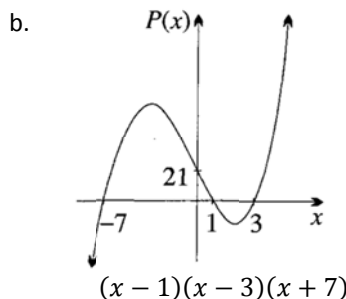
1. a. quotient: $x^2 - 3x + 5$, remainder: $12 - 13x$; b. $a = 8$ and $b = -5$; c. $(x^2 + x - 1)(x^2 - 3x + 5)$
2. a. $x^4 - x^3 + x^2 - x + 1 = (x^2 + 4)(x^2 - x - 3) + (3x + 13)$; b. $a = -4$ and $b = -12$; c. $(x^2 + 4)(x^2 - x - 3)$
3. $a = 41$ and $b = -14$
4. a. $k = 4$; b. $m = -\frac{1}{2}$; c. $p = -14$ d. $a = -1$
5. b. $x \geq 6$ or $-1 \leq x \leq 3$
6. b. $x \leq -2$ or $-\frac{1}{2} \leq x \leq 3$
- 7.



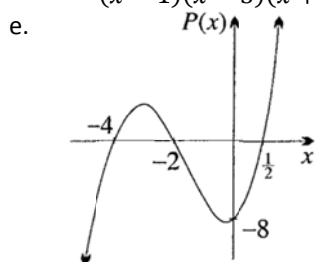
$$(x - 2)(x + 1)(x + 3)$$



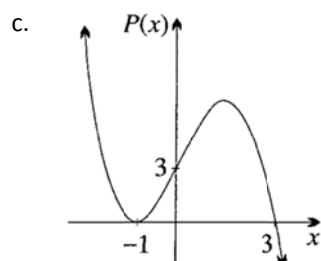
$$(x - 1)(x - 2)(x - 3)(x + 5)$$



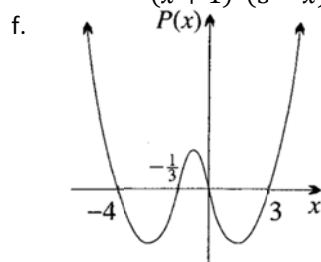
$$(x - 1)(x - 3)(x + 7)$$



$$(x + 2)(2x - 1)(x + 4)$$



$$(x + 1)^2(3 - x)$$



$$x(x - 3)(3x + 1)(x + 4)$$

8. a. $(2x - 1)(x + 3)(x - 2)$; b. $(3x + 2)(2x + 1)(x - 1)$
9. a. $P(x) = (x + 1)(x - 2)^2$, $Q(x) = (x + 1)(x - 2)(x + 3)$ and $R(x) = (x + 1)^2(x - 2)$; b. $(x + 1)(x - 2)$; c. $(x + 1)^2(x - 2)^2(x + 3)$
10. $1/4$, 1 and 3
11. a. $a = 4$ and $b = 11$; b. $a = 2$ and $b = -9$; c. $P(x) = -x^3 + 16x$; d. $p = 2$ or $p = 3$
12. a. $3 - x$; b. $3 - 2x$
13. a. $b = 0$, $c = -9$ and $d = 0$; c. $-3 < x < 0$ or $x > 3$
14. b. $t = 1$ or $t = -2$
15. a. $x + 1$ is a factor when n is odd; b. $x + a$ is a factor when n is odd.
16. a. $P(x) = (x - 1)(x + 3)Q(x) + (2x - 1)$; b. 1
17. a. $\frac{1}{2}$; b. $a = -2$ and $b = -7$; c. 8
18. b. $a = -2$
19. a. $a = 2$, $b = \frac{1}{3}$ and $c = \frac{5}{2}$; b. $a = -1$, $b = 3$, $c = \frac{1}{2}$ and $d = \frac{5}{4}$
20. a. $a = 3$, $b = -16$ and $c = 27$; b. $a = 2$, $b = -2$, $c = -7$ and $d = -7$; c. $(x + 1)^3 - (x + 1)^2 - 4(x + 1) + 5$; d. $a = 3$, $b = -2$ and $c = 1$
21. a. $2(x + 1)^2 - (x + 1) - 7$; b. $a = 2$, $b = 16$ and $c = 35$; c. $2(x - 2)^2 + 3(x - 2) + 1$
22. a. $(x + 1)^2 - 2(x + 1) + 1$; b. $(n - 4)^2 + 8(n - 4) + 16$; c. $(x + 2)^2 - 4(x + 2) + 4$
23. a. $a = 2$, $b = 1$, $c = -\frac{1}{2}$ or $a = 2$, $b = -\frac{1}{2}$, $c = 1$; b. $3(m - 1)^2 - 3(m - 2)^2 + (m - 3)^2$
24. a. $P(x) = (x - 2)^2(x + 5)$; b. $P(x) = (x - 1)(x + 3)(2x - 7)$
25. Prove